PROJECT SYNOPSIS

NEXUS POS: AN OFFLINE-FIRST DESKTOP POINT OF SALE & INTELLIGENT INVENTORY MANAGEMENT SYSTEM

Degree / Branch: B.Tech in Computer Science & Engineering

Academic Year: 2026 - 2027

Candidate Name: _____

Project Guide: Prof. _____

1. ABSTRACT / INTRODUCTION

Modern retail environments rely heavily on Software-as-a-Service (SaaS) cloud Point of Sale (POS) tools that enforce recurring subscription fees, continuous broadband internet connectivity, and external data storage. For independent retail merchants and local businesses, internet outages halt transaction queues, resulting in financial loss and dissatisfied shoppers.

Nexus POS is an offline-first desktop Point of Sale and inventory control system engineered to guarantee 100% operational uptime, zero subscription costs, and total data sovereignty. Developed using an **Electron, React 19, and SQLite3** architecture, the application delivers sub-10ms transactional processing, real-time stock deduction, cryptographic credential authentication, customer loyalty rewards, split-payment tracking, operational expense management, and dual-mode receipt generation (standard A4 and 80mm thermal receipt formats).

2. PROBLEM STATEMENT

Contemporary small-to-medium retail operations face critical barriers with traditional and cloud POS platforms:

- **Cloud Downtime Vulnerability:** Network interruptions disable register operations, causing lost sales and extended customer checkout wait times.
- **High Recurring Overhead:** Subscription-based models (Shopify POS, Square) demand continuous monthly platform and per-transaction fees.
- **Data Privacy & Security Risks:** Sensitive business revenue numbers, customer contact details, and product catalogs are stored on third-party remote servers.
- **Inventory Disconnection:** Manual stock spreadsheets create discrepancies between actual shelf counts and billed receipts.

3. OBJECTIVES

- Deliver an autonomous, zero-downtime offline transaction terminal utilizing an embedded SQLite database.
- Enforce Role-Based Access Control (RBAC) separating administrative controls from staff checkout terminals.
- Protect sensitive user credentials using memory-hard, salt-based cryptographic key derivation (`scrypt`).
- Support hardware barcode scanners through automated high-speed keystroke interception listeners.

- Provide client-side compilation of dual-format printable receipts (Formal A4 PDF and 80mm POS thermal rolls).
- Implement automated unit testing via Vitest and React Testing Library to guarantee stability.

4. SYSTEM ARCHITECTURE & TECHNICAL STACK

The system follows Electron's multi-process security architecture. The **Main Process (Node.js)** handles low-level file I/O, database transactions, and cryptographic routines. The **Renderer Process (React.js)** manages user interactions and visual presentation. Sandboxed communication is governed by an explicit **Inter-Process Communication (IPC)** bridge via `contextBridge`.

Layer	Technology	Role in System
Runtime Environment	Electron (v39.x / Node.js)	Cross-platform desktop container with native OS access
Frontend Framework	React 19 & Vite 7	Component-based reactive UI with rapid hot-module reload
Client Routing	React Router DOM (v7)	Single Page Application (SPA) client navigation
Embedded Database	SQLite3	ACID-compliant, zero-latency local persistent storage
Document Engine	jsPDF & jsPDF-AutoTable	Client-side dynamic compilation of A4 & Thermal bills
Cryptography	Node.js `crypto` (scrypt)	Salted password hashing with timing-safe validation
Automated QA	Vitest + RTL + JSDOM	Automated unit testing with mocked IPC bridges

5. KEY FUNCTIONAL MODULES

1. Authentication & Role-Based Access Control (RBAC): Multi-tier account system (Admin vs. Staff). Passwords are cryptographically salted and hashed using `scrypt` (stored format: `<salt>:<derivedKey>`). Staff members are restricted strictly to checkout screens.

2. POS Terminal & Hardware Barcode Integration: Split-screen catalog and active cart view. Implements rapid keystroke event listeners supporting standard USB barcode scanners with automatic submit on carriage return, split/partial payments (Cash/Card/UPI), and automatic change computation.

3. Inventory & Catalog Control: Comprehensive CRUD operations with SKU tracking. Automatically deducts stock upon bill completion and restores stock during invoice refunds with visual safety threshold alerts (Stock ≤ 5).

4. CRM & Customer Loyalty System: Customer directory searchable by phone number. Calculates and awards loyalty points (+1 point per currency unit spent) on every completed invoice.

5. Financial & Expense Tracking: Daily store expenditure ledger (rent, electricity, supplier payments) combined with gross revenues to compute accurate net operational performance.

6. Disaster Recovery & Local Backup: One-click local database backup creating clean filesystem-level duplicates of the SQLite database.

6. DATABASE SCHEMA (SQLITE)

Table	Primary Key	Key Attributes & Descriptions
users	`id` (INTEGER)	`username` (UNIQUE), `password` (scrypt hash), `role` (Admin/Staff)
products	`id` (INTEGER)	`name`, `sku` (UNIQUE), `price`, `stock`
invoices	`id` (INTEGER)	`customer_name`, `customer_phone`, `date`, `total`, `amount_paid`, `balance_due`, `payment_method`, `status`
invoice_items	`id` (INTEGER)	`invoice_id` (FK), `product_id` (FK), `quantity`, `price`
customers	`id` (INTEGER)	`name`, `phone` (UNIQUE), `points` (Loyalty rewards)
expenses	`id` (INTEGER)	`description`, `amount`, `date`
settings	`key` (TEXT)	`value` (Store name, currency symbol, tax percentage)

7. TESTING & QUALITY ASSURANCE

The application incorporates an automated testing pipeline configured with **Vitest** and **React Testing Library** running against an emulated JSDOM environment. Global mocks simulate Electron IPC endpoints, validating component lifecycles, user greetings, cart renders, and routing protections. All test suites pass with 100% success.

8. CONCLUSION & FUTURE SCOPE

Nexus POS demonstrates that modern desktop retail software can achieve the visual polish and ease-of-use of web apps while retaining the independence, zero-latency, and reliability of embedded local storage. Future enhancements include multi-terminal LAN synchronization over local WebSockets, raw ESC/POS thermal command generation for automated cash drawer triggers, and optional encrypted cloud backup mirrors.

Candidate Signature	**Project Guide**	**Head of Department**